

WEEK 2: FUNCTIONAL ANATOMY & KINESIOLOGY

MUSCLE ACTIVITY AND STRENGTH

SYSTEM	COMPONENTS	ROLE IN MOVEMENT
NMJ	CNS, peripheral nerves, muscles	Initiates, controls, and coordinates muscle forces for movement
Skeletal system	Bones and joints	Provides levers and joints that determine direction and ROM
Support and stabilizing structures	Ligaments, capsules, tendons, fascia	Maintains joint stability. transmits force, and supports controlled motions

MUSCLE AS A FORCE-PRODUCING TISSUE

- Functions of muscle:

- Produce movement
- Control movement
- Maintain posture
- Stabilize joints

- Muscle activity:

- Varies depending on the phase of movement and task requirements
 - Example: gait — quadriceps
 - Controls knee flexion during loading response
 - Contributes to knee stability during mid-stance

MOTOR UNITS

- Basic functional unit of muscle contraction; direct link between the nervous system and skeletal muscle

- Motor unit consists of:

- One alpha motor neuron
- All the muscle fibers innervated by that neuron

- All muscle fibers within a single motor unit

- Contract simultaneously
- Follow the all-or-none rule

FORCE REGULATION IN MUSCLE CONTRACTION

- The amount of force produced by a muscle does not depend on a single muscle fiber, but on how the nervous system controls motor units

- Force production depends on:

1.) Number of motor units activated — motor unit recruitment

- **Small motor units** - low force, fine control
- **Large motor units** - high force, gross movement
- Recruitment follows the size principle (small -> large)

2.) Frequency of motor neuron discharge — rate coding

- **Low firing rate** - twitch contraction
- **Higher firing rate** - summation
 - Cramps
- **Very high firing rate** - tetanic contraction (maximal force)

STRUCTURAL COMPONENTS OF THE SARCOMERE

1.) A band

- Contains myosin filaments
- Width remains constant during contraction and relaxation

2.) I band

- Contains only actin filaments
- Shortens during contraction
- Lengthens during relaxation or stretch

3.) H zone

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- Central region of the A band
- Contains only myosin
- Disappears during contraction

SUMMARY OF SLIDING FILAMENT THEORY

- Sarcomere - functional unit of skeletal muscle

Step	Event	Important Molecules	Result
1. Excitation	Action potential reaches muscle	ACh, Sarcolemma, T-tubules	Ca ²⁺ released from sarcoplasmic reticulum
2. Cross-Bridge Formation	Myosin heads attach to actin	Myosin, Actin, Troponin, Tropomyosin	Cross-bridge formed; ready for power stroke
3. Power Stroke	Myosin pulls actin toward sarcomere center	Myosin (ADP + Pi), Actin	Sarcomere shortens; muscle generates force
4. Detachment	Myosin releases actin	ATP	Myosin head resets for next cycle
5. Relaxation	Muscle stops contracting	Ca ²⁺ pumped back into SR, Tropomyosin covers actin	Muscle relaxes; sarcomere returns to resting length

TYPES OF SKELETAL MUSCLE FIBER

Type of Muscle Fiber	Type I	Type IIa	Type IIx
Other Names	Red fibers, Slow oxidative, Slow twitch	Fast oxidative	White fibers, Fast glycolytic, Fast twitch
Metabolism	Oxidative (aerobic)	Primarily aerobic	Anaerobic glycolysis
Fatigability	Low (≥ 30 seconds)	Moderate (12–30 seconds)	High (< 12 seconds)
Oxidative Capacity	High	High	Low
Substrate for ATP Production	Fatty acids	Glucose	Glucose
Fiber Size	Small	Moderate	Large
Myoglobin	High, more capillaries	Moderate	Low
Mitochondria	More	Moderate	Less

MUSCLE CONTRACTION TYPES

Type of Contraction	Muscle Length	Muscle Tension	Key Characteristics / Subtypes	Example
Isometric	Same	Variable	No joint movement	Plank, wall sit
Isotonic	Variable	Same	Produces movement	Lifting or lowering a weight
• Concentric	Shortens	Same	Muscle shortens while producing force	Biceps curl (lifting phase)
• Eccentric	Lengthens	Same	Muscle lengthens while under tension	Biceps curl (lowering phase)
Isokinetic	Variable	Variable	Velocity is constant (same speed)	Machine-based exercises (e.g., Cybex)

1.) Concentric - shortening

- Muscle's proximal and distal insertion move closer towards each other
- Positive work - force exerted by the muscle to produce motion

2.) Eccentric (negative work) - lengthening

- Negative work - outside force produces joint motion

3.) Isometric (static or holding) - no change in joint angle

- Stabilizes joints
- Early stage of rehab
- All of the muscle contracts

4.) Isotonic - same tone

- Only one muscle contracts, antagonists relaxes

5.) Isokinetic - same rate of movement or speed

E.g., Walking

- Foot strikes: concentric: tibialis anterior, quads, iliopsoas
- Mid-stance: isometric
- Rising onto the toes: isotonic

ROLE OF GRAVITY IN MUSCLE ACTIVITY

- Muscle function is influenced by the relationship between movement and gravity

- Muscle action may occur:

- Against gravity, with gravity, in a gravity eliminated position, in a neutral gravity plane

- This principle forms the basis for:

- MMT, exercise progression, functional movement analysis

LENGTH-TENSION RELATIONSHIP

- Describes the relationship a muscle's length and its ability to generate force

- Muscle generates maximum active resting length

- Factors affecting force production:

- Fiber type
- Number of muscle fibers
- % of fiber recruitment
- Fiber architecture
- Angle of pull
- Muscle temperature
- Rate of stimulation
- Force-velocity relationship
- Elastic properties